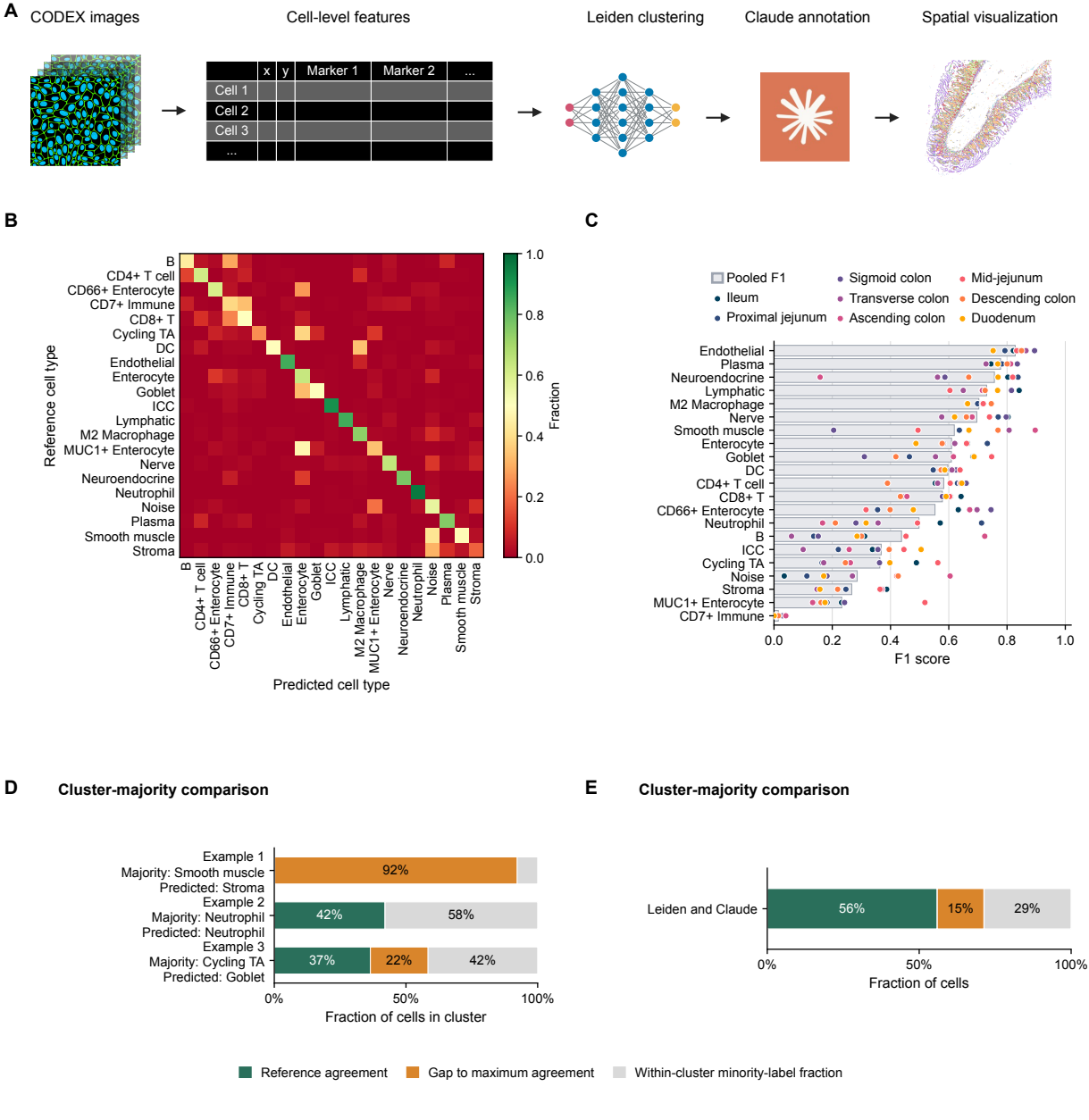

A
CODEX images
Cell-level features
Leiden clustering
Claude annotation
Spatial visualization
x
y
Marker 1
Marker 2
...
Cell 1
Cell 2
Cell 3
B
Reference cell type
Predicted cell type
Fraction
C
Pooled F1
Ileum
Proximal jejunum
Sigmoid colon
Transverse colon
Ascending colon
Mid-jejunum
Descending colon
Duodenum
F1 score
D
Cluster-majority comparison
Example 1
Majority: Smooth muscle
Predicted: Stroma
92%
Example 2
Majority: Neutrophil
Predicted: Neutrophil
42%
58%
Example 3
Majority: Cycling TA
Predicted: Goblet
37%
22%
42%
Fraction of cells in cluster
E
Cluster-majority comparison
Leiden and Claude
56%
15%
29%
Fraction of cells
Reference agreement
Gap to maximum agreement
Within-cluster minority-label fraction